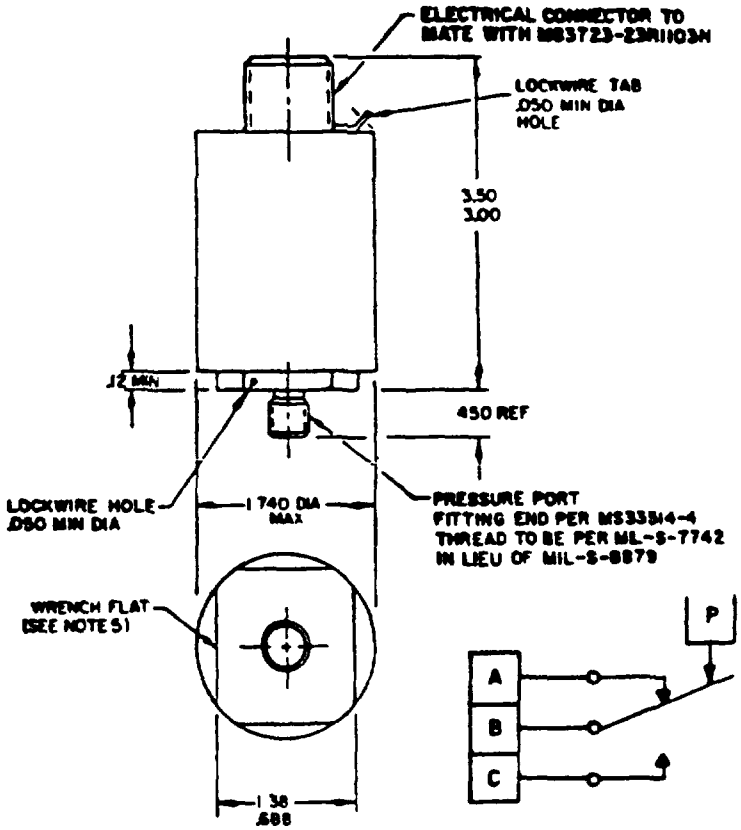
MIL-S-9395/30D
23 JULY 1980
SUPERSEDING
MIL-S-9395/30C
6 June 1977

MILITARY SPECIFICATION SHEET

SWITCHES, PRESSURE, (GAGE),
TYPE II, 5 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the pressure switches described herein shall consist of this document and the latest issue of Specification MIL-S-9395.



WEIGHT: NOT TO EXCEED .45 lbs.

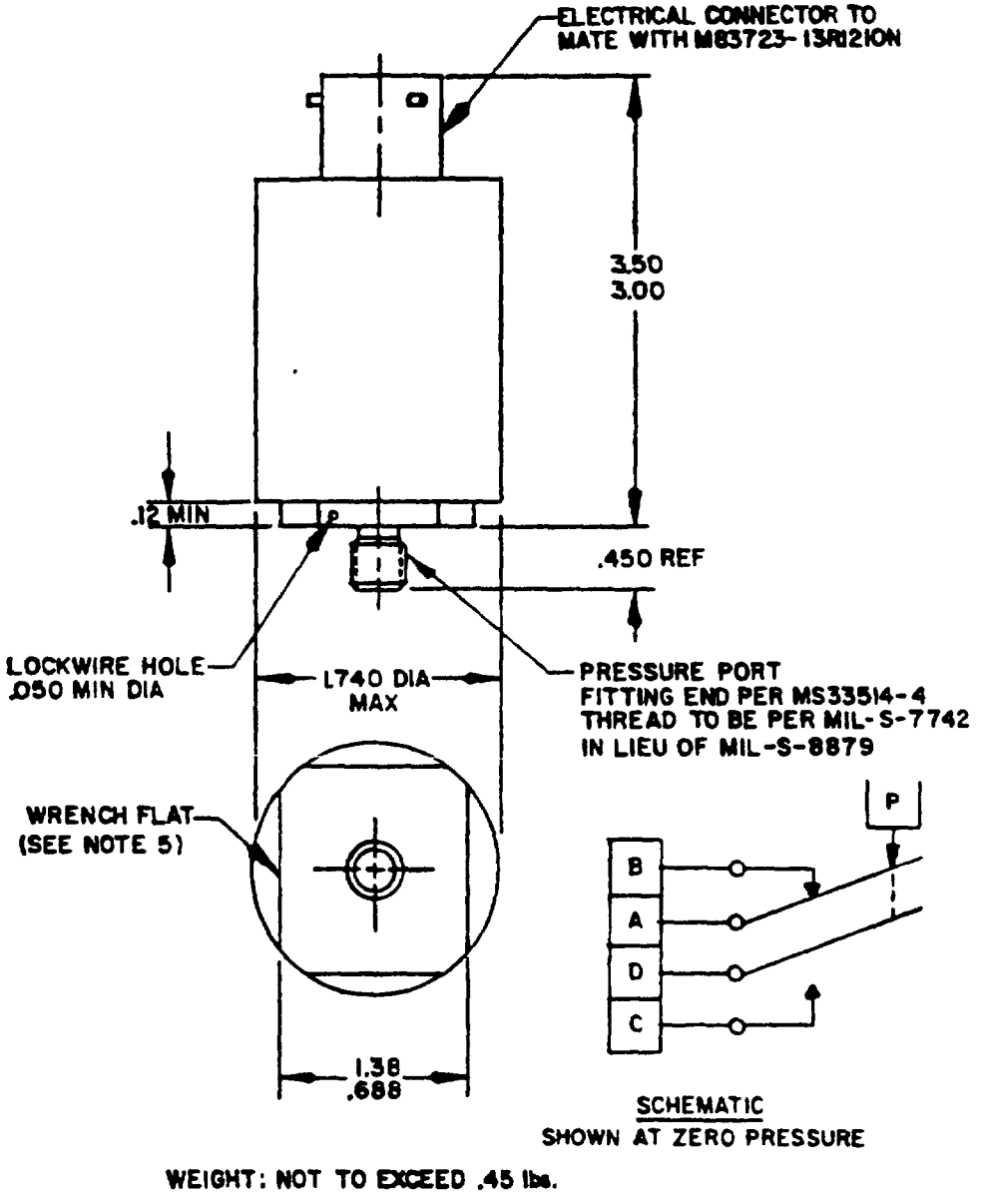
Configuration 1

FIGURE 1 Switches

(D) denotes changes.

FSC 5930

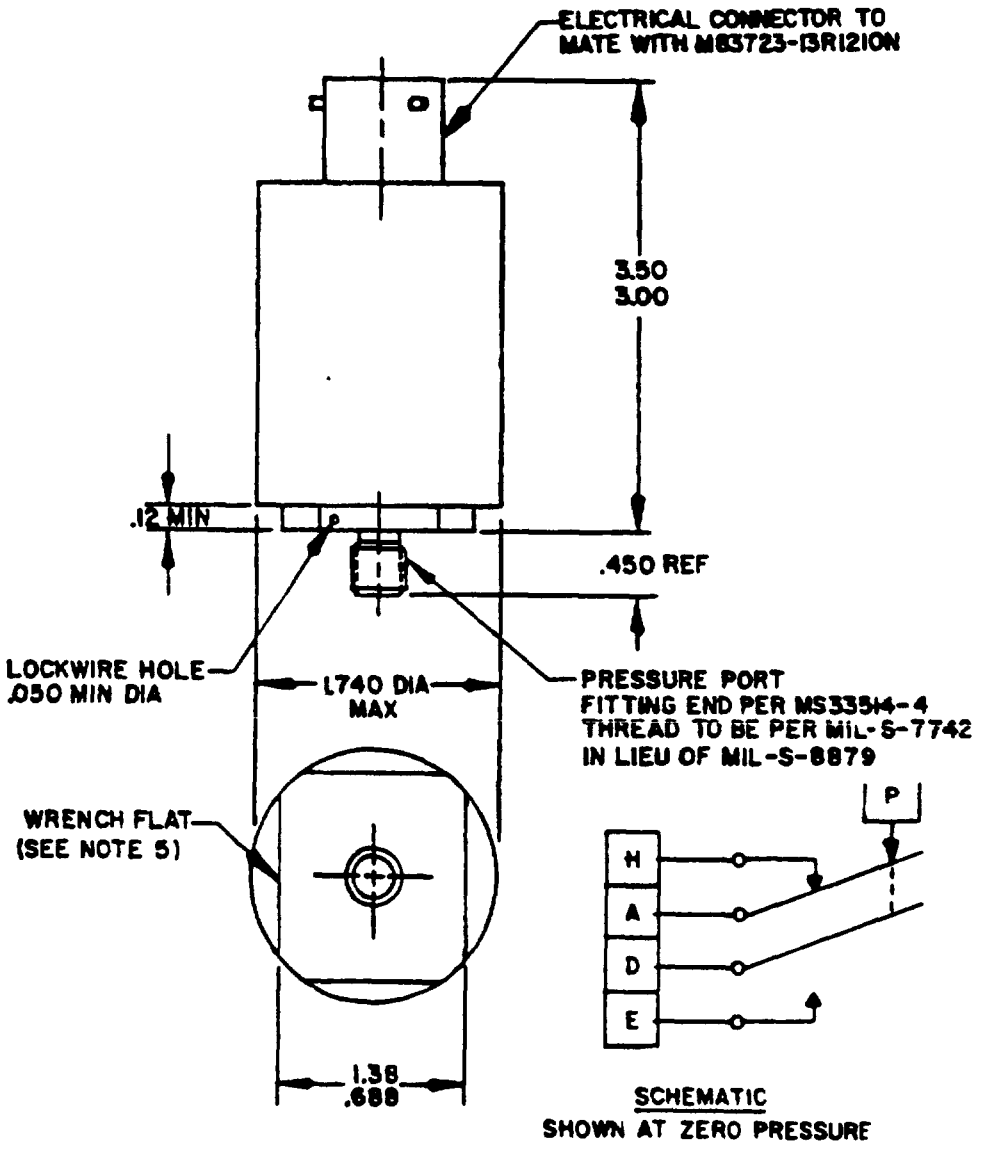
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Configuration 2

FIGURE 1. Switches - Continued.

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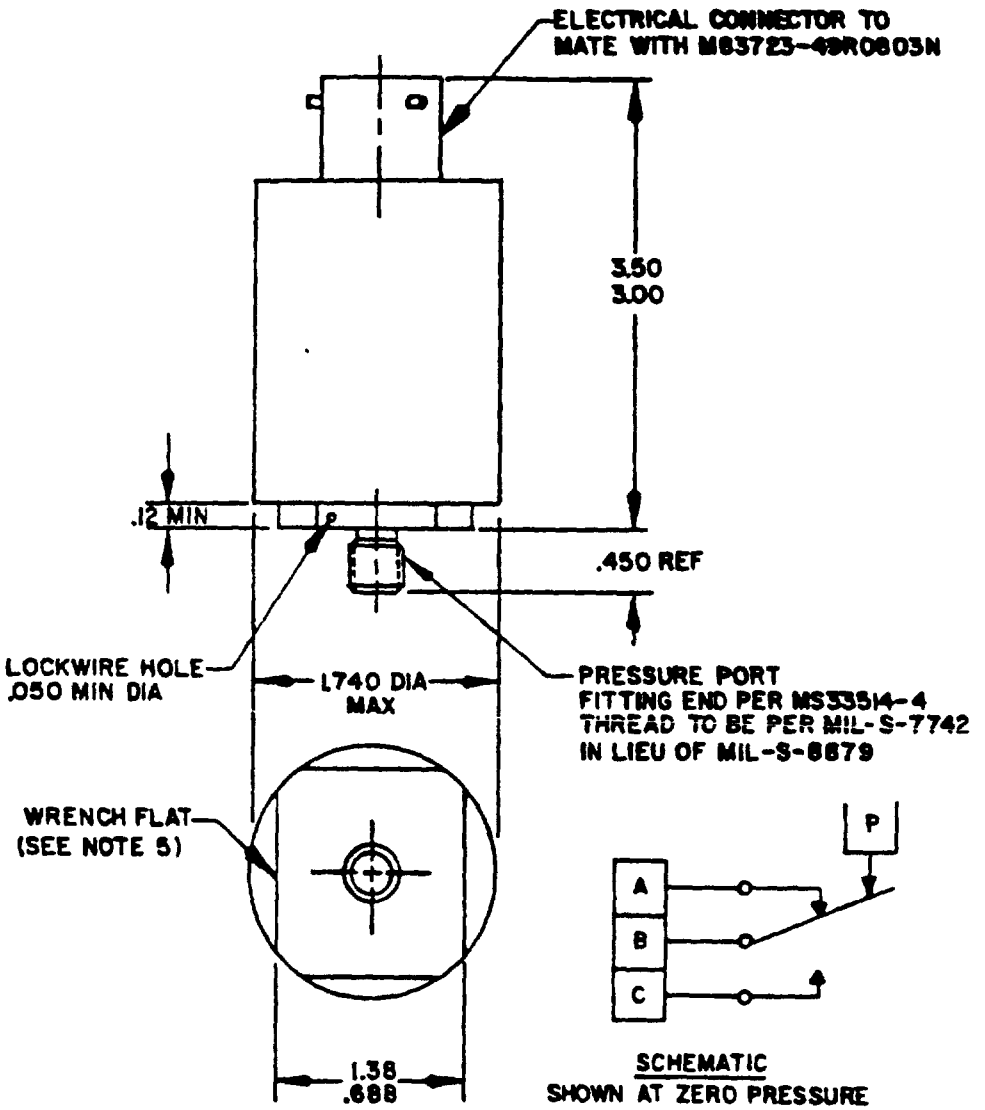


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 3

FIGURE 1. Switches - Continued.

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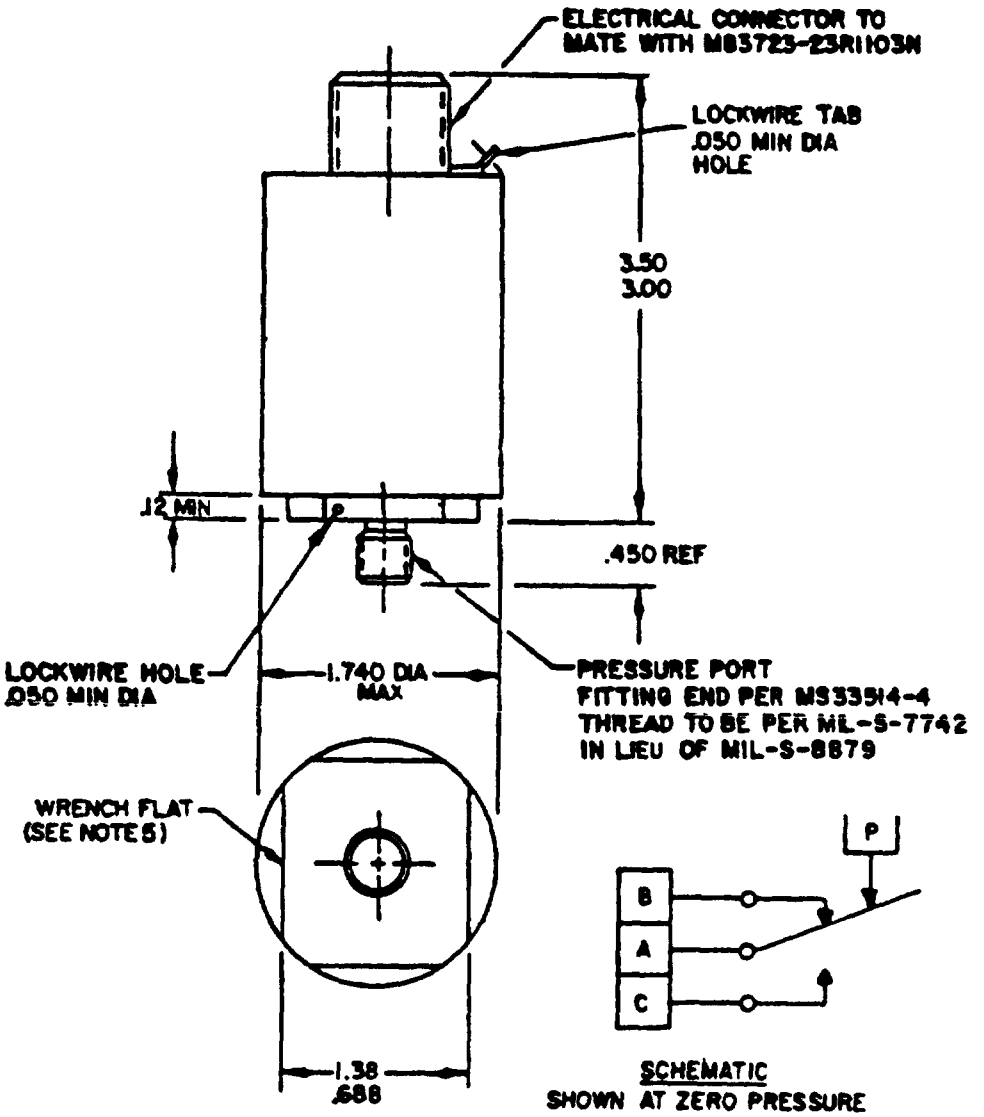


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 4

FIGURE 1. Switches - Continued.

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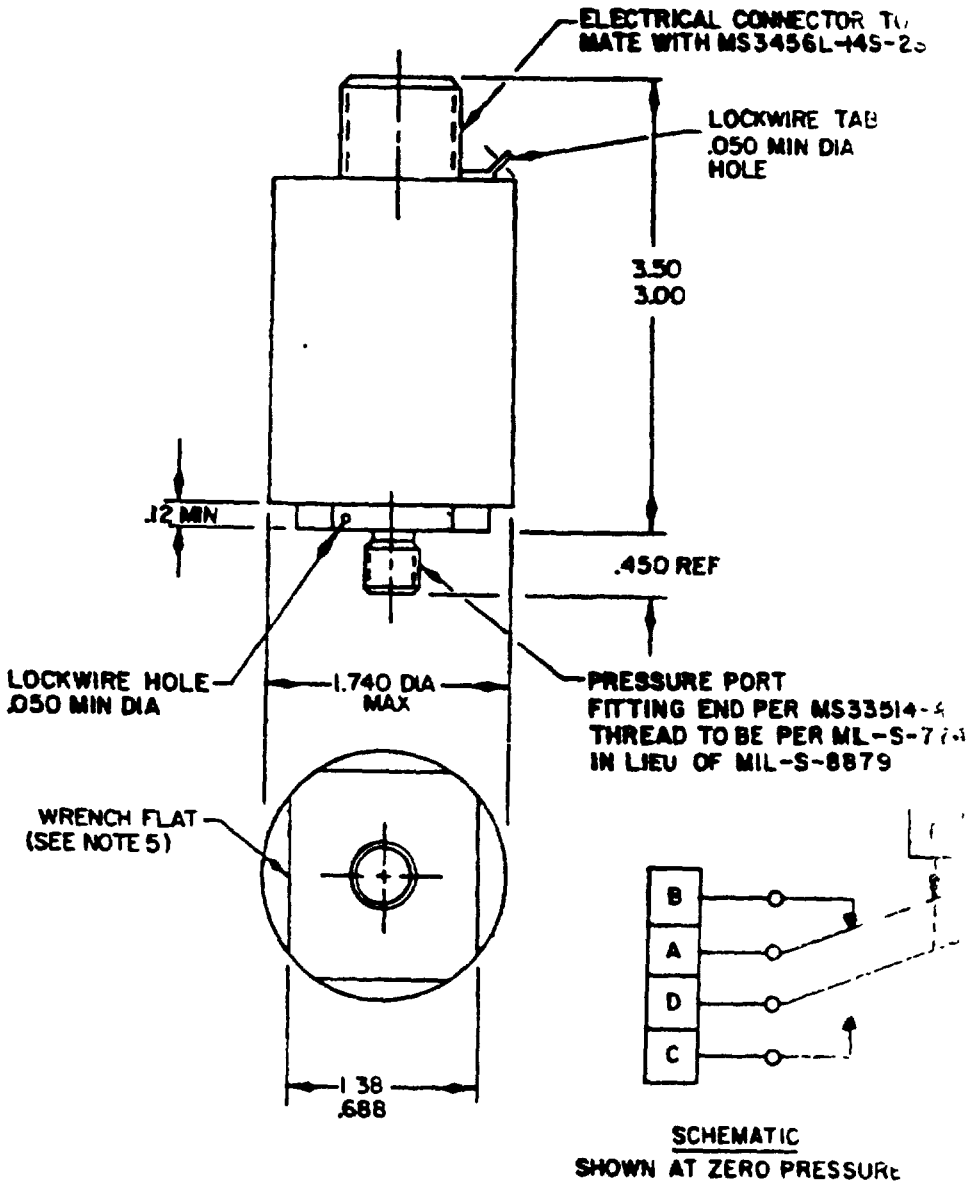


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 5

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
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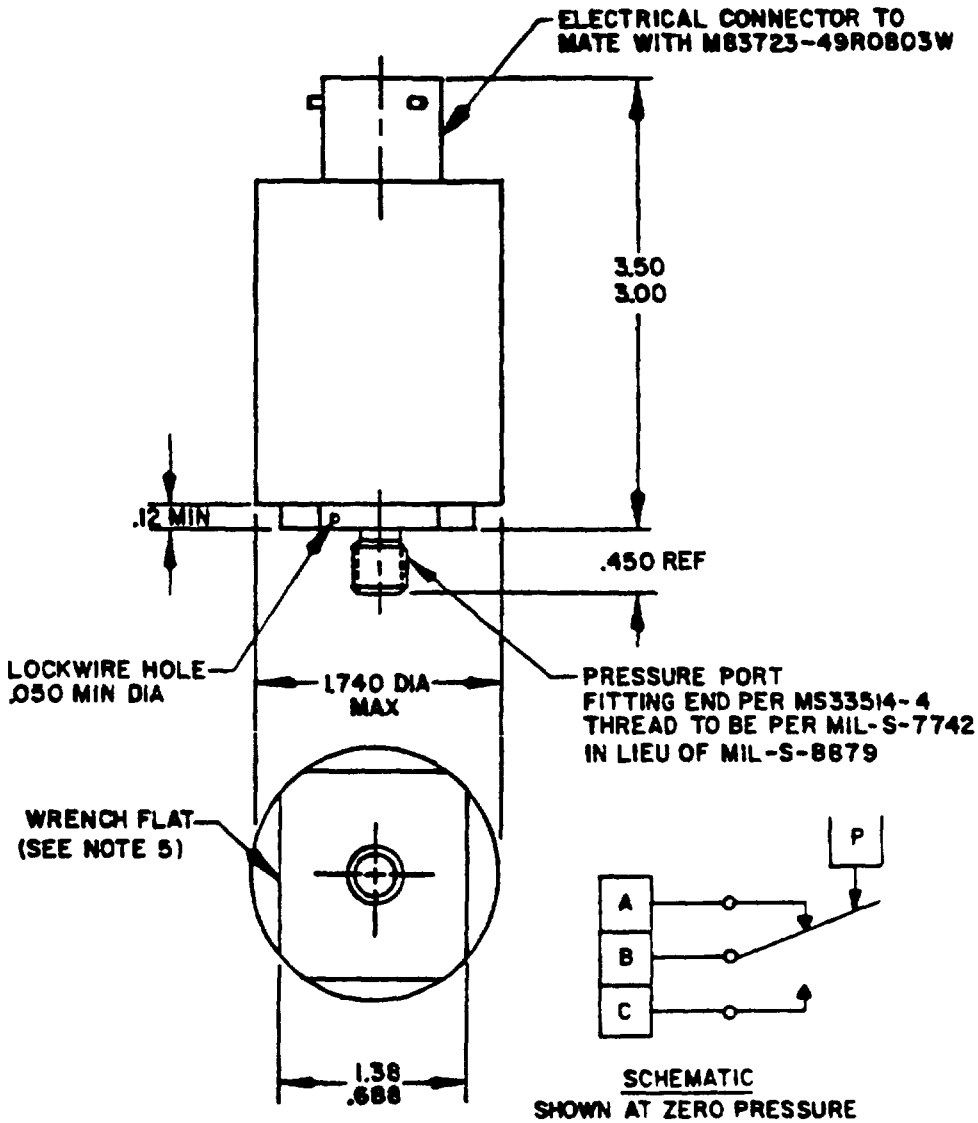


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 6

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
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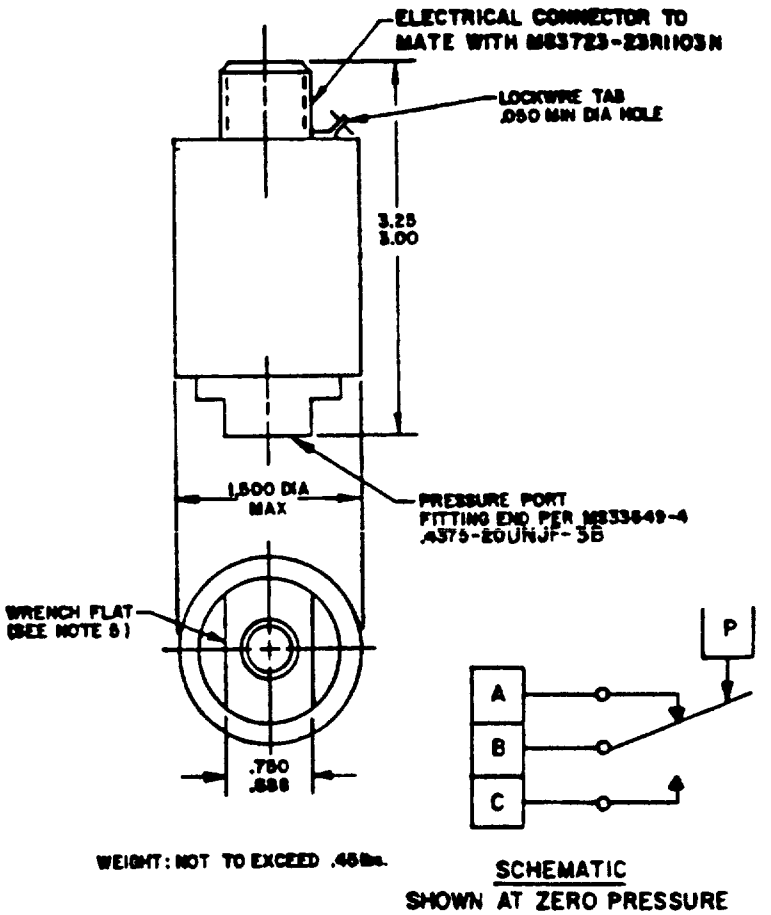


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 7

FIGURE 1. Switches - Continued.

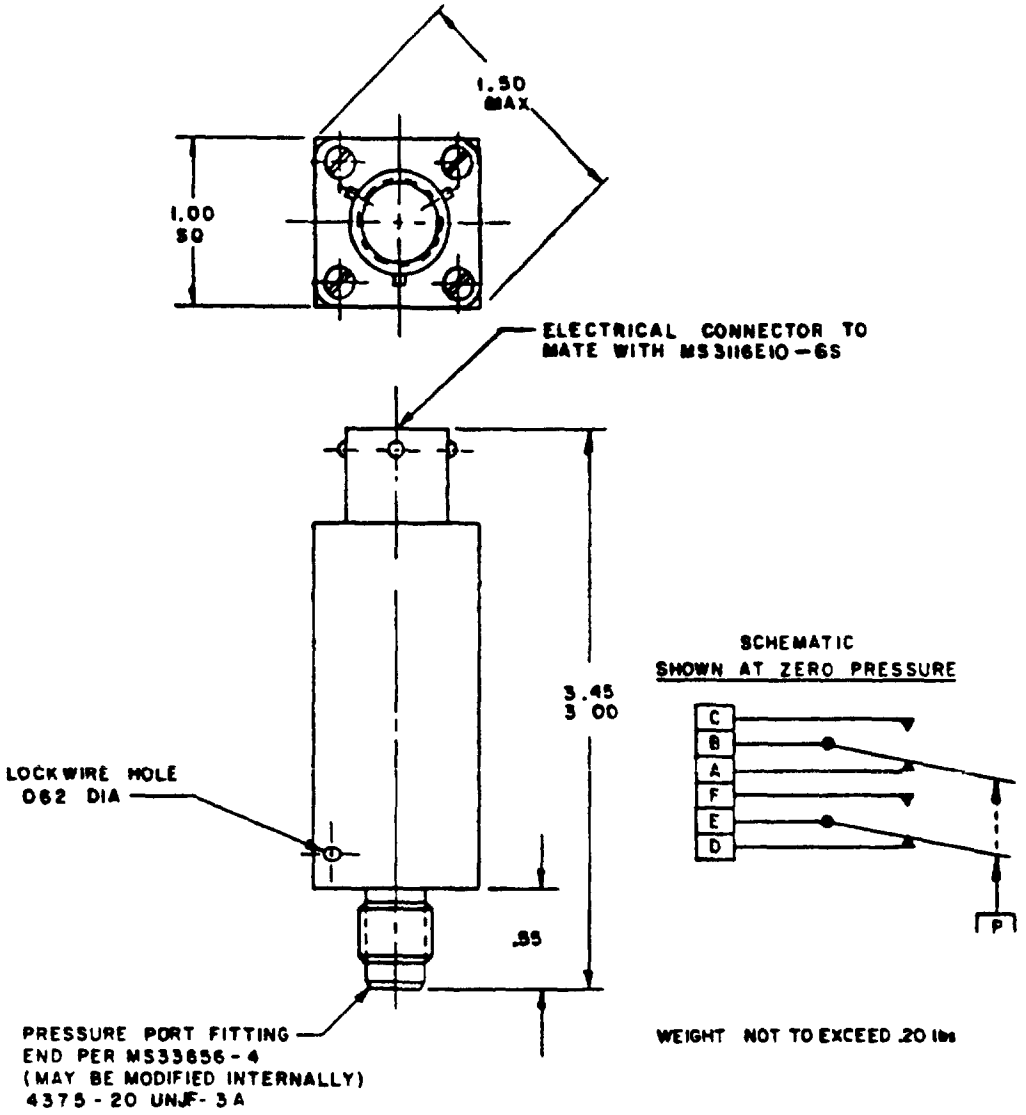
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
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Configuration 8

FIGURE 1. Switches - Continued.

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CONFIGURATION 9

FIGURE 1. Switches - Continued.

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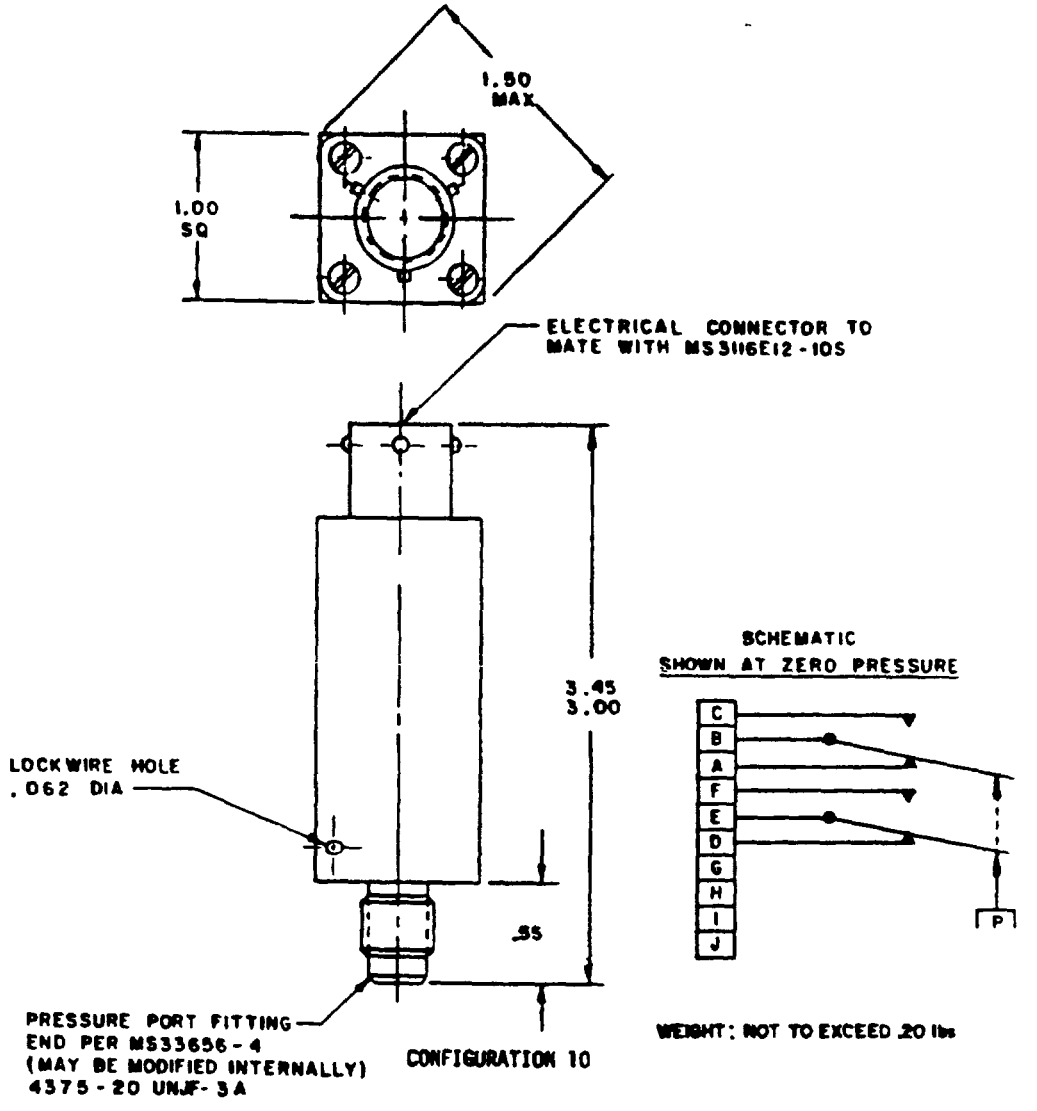


FIGURE 1. Switches - Continued.

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INCHES	MM
.005	.13
.050	1.27
.062	1.57
.12	3.0
.450	11.43
.55	14.0
.688	17.48
.750	19.05
1.00	25 4
1.38	35.1
1 50	38 1
1.740	44.20
3.00	76.2
3.45	87.6
3.50	88.9

NOTES
1. Dimensions are in inches.
2 Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm
3 Unless otherwise specified, tolerance is ±.005 (13 mm)
4 Exact shape of switch is optional provided outline dimensions specified are not exceeded and mounting holes and connector locations are as specified
5 A minimum of two wrench flats is needed

FIGURE 1. Switches - Continued

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REQUIREMENTS

Dimensions, weight, and electrical schematic: See figure 1.

Calibration See tables I, II, III, and IV.

Proof pressure: 4500 lb_f/in^2.

System pressure 3000 lb_f/in^2.

Burst pressure 7500 lb_f/in^2.

(D) Electrical ratings.
Operating voltage - 28 Vdc.
Current rating - 5 amperes inductive (L/R = .026).
2 amperes lamp load.

(D) Seal:
Pressure chamber: Media proof. Subject switches to proof pressure for 2 minutes using hydraulic fluid per MIL-H-5606 with chamber pressure continuously being monitored. Isolate the chamber at proof pressure, with the chamber disconnected from the pressure source. Under that condition, the pressure shall not drop more than 5 lb_f/in^2 for the first 30 seconds to allow for stabilization of test equipment. No pressure loss is allowed thereafter for the remainder of the 2 minutes.

Electrical chamber Watertight. In the event of primary seal failure, hydraulic fluid shall not enter the electrical chamber.

Electrical connector See figure 1.

Pressure port See figure 1.

Media Hydraulic fluid per MIL-H-5606.

High temperature (operating and nonoperating). B (+275°F).

Low temperature (operating and nonoperating): D (-65°F).

Altitude C (except 80,000 feet).

Shock C (100 G).

Vibration S (test condition C, method 204 of MIL-STD-202).

Life (mechanical) F (50,000 cycles).

Life (electrical). B (25,000 cycles).

Acceleration C (8 G).

Pulsation amplitude E (10 percent).

Pulsation frequency: D (500 ±50 Hz).

(D) Pressure rise F (500,000 $lb_f/in^2/sec$).

Dielectric withstanding voltage (at reduced barometric pressure): Applicable at 250 Vrms.

Connector torque Applicable.

QUALIFICATION

Single submission Restricted to switch submitted.
Group submission See table V.

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(D) PART NUMBER Consists of the prefix "M9395/30-" followed by a five-character code. The code identifies the configuration and pressure setting mode (code from table I), high-pressure setting to within 100 lb_f/in^2 (code from table II) followed by high-pressure setting to within 25 lb_f/in^2 and applicable tolerance (code from table III), and low-pressure setting to within 100 lb_f/in^2 (code from table II) followed by low-pressure setting to within 25 lb_f/in^2 and applicable tolerance (code from table III). The five-character code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 300 $\pm$25 lb_f/in^2, and deactuates on decreasing pressure at 150 $\pm$25 lb_f/in^2.

EXAMPLE

M9395/30- A CA AC

(D) PREFIX. M9395/30-

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I. A

HIGH-PRESSURE SETTING = Code "C" of table II (300 lb_f/in^2) and Code "A" of table III (0, $\pm$25 lb_f/in^2). CA

LOW-PRESSURE SETTING = Code "A" of table II (100 lb_f/in^2) and Code "C" of table III (+50, $\pm$25 lb_f/in^2). AC

TABLE I Codes for combinations of configurations and pressure setting modes.

	Configuration										Pressure setting mode			
	1	2	3	4	5	6	7	8	9	10	High setting		Low setting	
Code- - -	A	D	G	K	N	R	U	X	1	4	At (or max)	1/	At (or min)	1/
Code- - -	B	E	H	L	P	S	V	Y	2	5	At (or max)	1/	Differential	2/
Code- - -	C	F	J	M	Q	T	W	Z	3	6	Differential	2/	At (or min)	1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

TABLE II. Codes for pressure settings to within 100 lb_f/in^2.

Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)
A - - -	100	L - - -	1100	W - - -	2100
B - - -	200	M - - -	1200	X - - -	2200
C - - -	300	N - - -	1300	Y - - -	2300
D - - -	400	P - - -	1400	Z - - -	2400
E - - -	500	Q - - -	1500	1 - - -	2500
F - - -	600	R - - -	1600	2 - - -	2600
G - - -	700	S - - -	1700	3 - - -	2700
H - - -	800	T - - -	1800	4 - - -	2800
J - - -	900	U - - -	1900	5 - - -	2900
K - - -	1000	V - - -	2000	6 - - -	3000

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TABLE III. Codes for pressure settings to within 25 lb_f/in^2 and tolerances.

	Unit				Tolerance (lb_f/in^2)
	0	+25	+50	+75	
Code - - -	A	B	C	D	±25
Code - - -	E	F	G	H	±50
Code - - -	J	K	L	M	±75
Code - - -	N	P	Q	R	±100
Code - - -	S	T	U	V	±150
Code - - -	W	X	Y	Z	±200
Code - - -	1	2	3	4	±300
Code - - -	5	6	7	8	±400
Code - - -	9	0	I	-	Min or Max

TABLE IV. Codes for differential pressure settings.

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	M - - -	275
B - - -	25	N - - -	300
C - - -	50	P - - -	325
D - - -	75	Q - - -	350
E - - -	100	R - - -	375
F - - -	125	S - - -	400
G - - -	150	T - - -	425
H - - -	175	U - - -	450
J - - -	200	V - - -	475
K - - -	225	W - - -	500
L - - -	250		

(D) TABLE V. Extent of qualification.

Part number	Number of samples required	Tests	Qualifies
M9395/30-AADAA	1 each inductive	Complete per qualification inspection of MIL-S-9395	All configurations
-KADAA	1 each lamp		
-14ZXV	1 each inductive		
-44ZXV	1 each lamp		

NOTES

1. Pressure switches supplied to this specification sheet shall be designed to operate normally with the system pump ripple at 3000 lb_f/in^2 at 0 to 700 Hz and average transient of 3000 ±500 lb_f/in^2 at 10 Hz peaking to 4050 lf_f/in^2.
2. Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number. The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or procured

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Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - MC, AS, SH
Air Force - 19

Preparing activity:
Air Force - 85

(Project 5930-1113-4)

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POSTAGE AND FEES PAID
DEPARTMENT OF AIR FORCE
DOD-318

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

AFIT/DPPS-Stin
[illegible], OH 45433

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